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연구 제목	Neural Differentiation of Primary and Secondary Reward Processing in Addiction : A Systematic Review

1. Background and significance of study

Addiction and decision-making are closely intertwined, as individuals repeatedly make choices regarding whether to engage in behaviors such as eating, taking drugs, or gambling. Value-based decision-making is typically governed by two core components—reward and risk [16]. Accordingly, addiction has been conceptualized as a chronic brain disease involving disruptions in reward processing and motivational systems [31].

In the context of addiction, reward functions analogously to positive reinforcement: it is a stimulus that increases the likelihood of a behavior while inducing positive hedonic effects [17]. Depending on the addiction type, stimuli such as drugs, food, sex, or money may serve as rewards, evoking both "liking" and "wanting." These reward-driven responses can lead to habitual and compulsive engagement with a specific class of rewarding stimuli.

These rewards are commonly categorized into primary rewards (e.g., food, sex) and secondary rewards (e.g., money, social approval). Primary rewards are biologically essential and possess intrinsic value, whereas secondary rewards acquire value through social and associative learning [24]. Beyond these traditional categories, higher-level cognitive rewards—rooted in abstract cognitive representations—have also been recognized as key drivers of motivated behavior [23].

Both primary and secondary rewards act as potent behavioral reinforcers and can contribute to the development and maintenance of addiction. For instance, drug addiction and obesity both exhibit heightened salience attribution to specific rewards (drugs or food) and share impairments in dopaminergic pathways [30]. The ventral striatum, including the nucleus accumbens, is central to the processing of drugs of abuse [32]. Similarly, gambling disorder—primarily linked to money as a secondary reward—shows striatal hyperactivation patterns that parallel those observed in substance addiction [14].

While substantial evidence indicates that primary and secondary rewards share common underlying neural mechanisms in addiction, existing research often treats these reward types independently. A more integrative framework that accounts for how different reward categories differentially and similarly influence addiction is necessary to refine our understanding of reward-driven maladaptive behaviors. Therefore, the present study aims to synthesize neural evidence on reward processing across diverse forms of

addiction—including substance and behavioral addictions— to understand reward mechanisms in addiction in an integrated way.

One of the primary challenges of this review is the need to critically integrate findings from two largely independent studies, those examining primary rewards and those examining secondary rewards. Although each line of research has contributed important insights into addiction and reward processing, a unified framework is necessary to clarify how different reward types shape addictive behaviors and their underlying neural mechanisms. Building on this synthesis, this review aims to identify key gaps in the literature and propose future research directions that can further validate and expand a comprehensive framework of reward processing in addiction. Accordingly, this review addresses the following central research question: "What are the common and differential neural responses to primary and secondary reward stimuli within the reward circuit among individuals with substance or behavioral addictions?"

2. Research methods and results

This systematic review was conducted under the 2020 Preferred Reporting Items for Systematic Reviews (PRISMA) guidelines [21].

1) Search Strategy

The literature was searched in the PubMed database. No restriction was placed on the publication date of the previous studies, and all eligible literature published up to December 2025 was included. The search strategy included the following terms using a wildcard search: ("primary reward" OR "sexual reward" OR "food reward" OR food OR erotic) OR ("secondary reward" OR "monetary reward" OR "financial incentive" OR "social approval" OR reputation OR money) OR ("drug reward")) AND ("addiction" OR "substance use disorder" OR "behavioral addiction" OR "substance use" OR "substance dependence" OR gambling OR gaming OR "internet gaming disorder") AND ("neuroimaging" OR fMRI OR PET OR "reward circuitry" OR striatum OR dopamine) AND (humans[MeSH Terms]) AND (english[lang]). The categorization of primary and secondary rewards was based on Sescousse et al. (2013), with the inclusion of drug stimuli as an additional primary reward. Using Rayyan, the following tags were set and used to classify the relevant papers: Tag A: primary reward-related addiction, and Tag B: secondary reward-related addiction.

2) Inclusion and Exclusion Criteria

- Inclusion

The studies included in this review had to meet the following criteria: publication in English; inclusion of a study population involving individuals with substance use disorder, behavioral addiction, or healthy control participants; and the reporting of neurological outcomes such as fMRI, PET, or EEG-based brain activity, specifically focusing on reward circuit responses. Furthermore, the study design was restricted to experimental tasks (e.g., incentive delay tasks) that involved reward stimuli (primary or secondary rewards) and simultaneous utilization of brain imaging techniques.

- Exclusion

Studies were excluded if they involved only animal studies; reported a wrong outcome, meaning only behavioral results without brain or reward-related neural indicators; featured wrong categorization, specifically failing to properly indicate primary and secondary rewards; or represented the following publication type (e.g., book chapters, reviews, conference abstracts, dissertations, study protocols, pilot studies, case reports or opinion papers).

3) Data Extraction and Synthesis

The extracted data included the first author, publication year, population, sample size, task, contrast, modality and findings with relevant results (e.g., neural results and p-values). The extracted data are synthesized in the Results sections. Table 1 summarizes addiction studies related to primary rewards, while Table 2 summarizes those related to secondary rewards. The structure of both tables is consistent: the Sample (N) column provides details on the number of participants, the type of addiction, and the diagnostic criteria (e.g., DSM-IV, YFAS). The Task column outlines the specific type of task employed in each study (e.g., Imaging task, Monetary Incentive Delay Task) and the reward types utilized within the task (e.g., Monetary cues, Drug cues, Food cues). The Contrast column details the main items contrasted within the research and the Modality column indicates the neuroimaging technology used in each study (e.g., fMRI, PET). Lastly, the findings were categorized as primary if they were directly related to neural activity during reward processing, and as secondary if they provided supplementary explanations or context to the primary findings.

4) Results

- General Characteristics of the Included Studies

A total of 22 studies dealing with neural circuit of reward processing in addiction were included in this review. Table 1 summarizes 12 studies of primary reward: food, drug, sex. Table 2 summarizes 10 studies of secondary reward: money. The addiction types associated with primary rewards included substance use disorder (SUD) such as cocaine (CUD), alcohol (AUD), heroin (HUD), as well as food addiction. The addiction type related to money as the secondary reward, included data from pathological gamblers.

Based on the synthesis of the 22 studies, the median total sample size was 35, and the mean was approximately 39.82. The sample size ranged from a minimum of 7 participants [11] to a maximum of 99 participants [15]. Primarily, fMRI and PET neuroimaging techniques were used. The tasks included Cue-Induced Craving tasks, Monetary Incentive Delay Tasks (MIDT), Conditioning and Extinction tasks, and Probabilistic Reversal Learning Tasks (PRLT).

- Shared Neural Signatures

Studies on addictions related to primary rewards (substances, food, and sexual stimuli) and secondary rewards (money) commonly suggest that the striatum—a core region of the reward-processing circuit regardless of reward types [13]— and the prefrontal cortex (PFC) are jointly involved in reward processing. Studies consistently reported that responses to reward cues and the processing of craving are closely associated with striatal activation in individuals with both primary reward addictions (PA) and secondary reward addictions (SA) [1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 15, 19, 22, 24, 26, 28, 29, 30, 33]. Specifically, individuals with PA show strong reactivity to addiction-related cues [15]. For example, exposure to smoking cues leads to significant increases in blood flow in the ventral striatum [9], and individuals with HUD exhibit hyperactivation in the left nucleus accumbens when exposed to drug cues [12]. Consistent with these findings, individuals with SA—such as those with problem gambling (PG)—also show typical activation patterns in the nucleus accumbens during reward processing (Win vs. Lose) [19] and exhibit stronger activation in the bilateral ventral striatum when anticipating larger monetary rewards [28].

Activation patterns in the orbitofrontal cortex (OFC) and other prefrontal regions have also been observed. In PA, stimuli such as smoking cues [4, 9], anticipation of food rewards [10], and drug craving [12] elicit activation in the vmPFC and OFC, and activation of the dorsolateral prefrontal cortex (DLPFC) increases particularly in individuals with high

levels of food addiction. Similarly, in SA, greater prefrontal activation has been reported during gambling tasks [11]. In particular, among individuals with problem gambling, the posterior OFC shows heightened responsibility to monetary rewards during the outcome phase [24]. This is consistent with previous research on the common neural mechanisms of rewards [25, 27].

- Distinct Neural Signatures

In PA, amygdala activation is clearly observed during cue exposure, whereas in SA, amygdala activation during the reward anticipation phase shows a more complex pattern, including an inverse relationship with addiction severity. More specifically, in PA, the amygdala is activated during cue exposure or food-reward anticipation, reflecting its role in emotional salience and hedonic processing [4, 8, 9, 10]. In contrast, in SA, among individuals with problematic gambling (PG), activation of the right amygdala showed a significant negative correlation with gambling severity [28]. Additionally, for monetary reward stimuli, responses of amygdala to the addiction-related cues themselves tended to be less pronounced. These observations support the established distinction between the neural mechanisms of primary and secondary rewards. In particular, they align with findings that primary rewards elicit prominent amygdala activation, reflecting their sensory-driven and automatic processing characteristics [25].

Both types of addiction showed impaired connectivity within the prefrontal–striatal circuitry, but the specific regions affected differed across groups. For example, in food addiction, FA severity was associated with impaired functional connectivity between the anterior cingulate cortex (ACC) and striatal regions including the left caudate and left nucleus accumbens [6]. In contrast, in gambling addiction, the typical positive coupling of BOLD responses between the dorsal and ventral striatum observed in healthy controls was absent during exposure to gambling cues, indicating disrupted integration within striatal subregions [2].

Additionally, this review indicates differences in prefrontal responses across addiction types. In heroin use disorder, drug cues evoke hyperactivation in the right OFC and vmPFC, with vmPFC activity positively related to craving [12]. In contrast, individuals with problem gambling show significantly greater prefrontal activation—especially in right anterior and dorsolateral regions—when gambling for money compared to points [11]. Therefore, while both shows altered prefrontal engagement, PA involves heightened valuation-related vmPFC/OFC responses, whereas SA shows stronger activation in decision-making–related prefrontal areas.

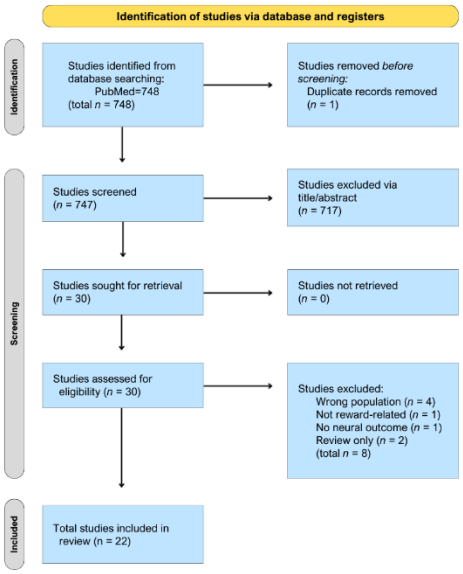
3. Meaning of Research

This systematic review is significant in that it clarifies the commonalities and differences in the processing mechanisms of primary and secondary rewards within the context of addiction by systematically reviewing the neural processing of rewards associated with various addiction types. In particular, this review confirmed the common role of the striatum in reward processing, while highlighting the differentiated responses of the amygdala and prefrontal cortex. The review holds further significance in that it also marginally identified differences in the impairment of prefrontal–striatal circuitry among individuals in different addiction types.

However, the review of existing studies identified a critical limitation: the need for more realistic fMRI experimental designs that provide actual rewards to individuals with addiction. For instance, in the case of nicotine addiction, more naturalistic neural activation patterns might be observed when actual nicotine rewards are administered, rather than merely presenting cigarette-related cues [3, 9, 26]. It is hoped that future neurological research on reward processing mechanisms across various types of addiction will advance through such naturalistic studies, grounded in a deeper understanding of the convergent yet divergent neural activities associated with primary and secondary rewards.

Figures

Figure 1. PRISMA flow Diagram



Tables

Table 1

Authors (Year)	Samples (N)	Task	Contrast	Modality	Primary findings	Secondary findings
Franklin et al. (2007)	Smoker (n=21) *Diagnosed with nicotine dependence according to DSM-4 criteria	Cigarette Cue-Induced Craving (CIC)	Smoking Stimuli vs. Non-smoking Stimuli	fMRI	Blood flow was significantly greater during exposure to smoking stimuli compared to nonsmoking stimuli in following area : Ventral striatum, Amygdala, Orbitofrontal cortex, Hippocampus, Medial thalamus, Left insula (p<0.01)	Perfusion positively correlated with the intensity of subjective cigarette *CIC in two main regions: Dorsolateral Prefrontal Cortex (r^2=0.54), Posterior Cingulate Cortex (r^2=0.53) (p<0.05) *Cigarette Cue-Induced Craving
Volkow et al. (2006)	Cocaine-addicted subjects (n=18) *Diagnosed with DSM-4 criteria for cocaine dependence and active users for at least 6 months	Watching a Cocaine-Cue Video	Neutral Video (non-repeating nature scenes)	PET	DV was significantly lower in the cocaine cue (1.53 ± 0.26, 1.37 ± 0.24) than in the neutral cue (1.62 ± 0.22, 1.44 ± 0.20) in the Putamen (p<0.05) and a trend in Caudate (p<0.06) indicating that the cocaine cues triggered a significant increase in DA release in the Dorsal striatum	(1) Cocaine cue significantly increased *CCQ scores (2.9 ± 1.4 to 3.5 ± 1.4) (p<0.01) *CCQ=Cocaine Craving Questionnaire (2) Correlation analysis between the DA changes and the clinical scales showed a significant association with the *CSSA in the Caudate (r=0.55) (p<0.01) and *ASI in right Putamen (r=0.47; p<0.05), left and right Ventral striatum (r=0.50) (p<0.04) *CSSA=Cocaine Selective Severity Assessment Scale *ASI=Addiction Severity Index
Bühler et al. (2010)	DS (n=21) OS (n=21) *DS=Dependent smokers OS=Occasional smokers *Diagnosed with DSM-4 criteria and FTND (Fagerström Test for Nicotine Dependence)	Event-related instrumental motivation task with Monetary and Cigarettes reward	Between-group comparison	fMRI	Anticipatory Brain Activity: - OS: Significantly higher activity in monetary reward compared to cigarettes reward - DS: No significant difference in inferior orbitofrontal gyrus, medial superior frontal gyrus, anterior cingulate gyrus, globus pallidus, putamen including caudate nucleus and ventral striatum (p<0.001)	Anticipatory Brain Activity and Subsequent Instrumental Responding: Increased activity in the caudate nucleus, putamen and ventral striatum during the anticipation phase were significantly linked to subsequently increased instrumental responding behavior (p<0.001)
Gearhardt et al. (2011)	High FA score (n=15) Low FA score (n=11) *Diagnosed with Yale Food Addiction Scale (YFAS)	The milkshake paradigm (1) Anticipated consumption (cue) (2) Real consumption	Between-group comparison	fMRI	(1) Anticipatory Food Reward : The High FA showed greater activation in Dorsolateral Prefrontal Cortex and Caudate nucleus compared to the Low FA (p=0.004) (2) Consummatory Food Reward : The Low FA showed greater activation in Lateral OFC compared to the High FA (p=0.009)	(1) Anticipatory Food Reward: Activation in ACC, Medial OFC and Amygdala in response to the anticipated receipt of palatable food compared to control cue (2) Consummatory Food Reward: no significant relationship
Dodds et al. (2012)	Overweight and obese participants with binge eating and emotional eating behavior (n=26) *Diagnosed with Y-BOCS-BE(Q6) and TFEQ-R18(Q3, Q6 and Q10) questionnaire	Imaging task	(1) Food images vs. non-food images (2) High calorie food images vs. Low calorie food images	fMRI	(1) Food > Non-food : Significantly greater activation in Amygdala, Caudate, Nucleus accumbens and Putamen (p<0.05) (2) High calorie > Low calorie : Significantly greater activation in Amygdala, Insula, Nucleus accumbens and Putamen (p<0.05)	-
Cambridge et al. (2013)	Moderate binge eating participants (n=30) *Diagnosed with Binge Eating Scale (BES)	Imaging task	High-calorie foods vs. Low-calorie foods vs. Rewarding nonfood items vs. Less rewarding nonfood items	fMRI	High calorie > Low calorie : Significantly higher activation in nucleus accumbens, putamen, midbrain, prefrontal cortex, insula, hypothalamus, and amygdala	Changes in mu-opioid receptor function influence stimulus-specific putamen and globus pallidus responses in obese patients with binge eating tendencies (p<0.05)
Sweitzer et al. (2014)	Daily Smokers (n=38)	Rewarded Guessing Task with Monetary and Cigarette reward cues	Abstinent Condition vs. Non-abstinent Condition	fMRI	Abstinence condition (1) Significantly higher activation in the bilateral caudate nucleus and the medial prefrontal cortex when anticipating a cigarette reward than others (p<0.001) (2) Significantly lower activation in the bilateral caudate when anticipating a monetary reward than others (p<0.01)	Participants who showed a greater decrease in activation to monetary reward anticipation due to abstinence also exhibited (1) a greater increase in *craving - left caudate nucleus (r=0.322, p<0.05) and mPFC (r=0.433, p<0.01) (2) an increase in *withdrawal symptoms - bilateral caudate nucleus (r=0.353, p<0.05) *Measured by 4-item Questionnaire on Smoking Urges scores *Measured by Minnesota Withdrawal Scale
Koban et al. (2023)	*SUD (n=59) HC (n=40) *SUD=Substance Using Disorder, HC=Healthy Controls *Diagnosed with DSM-4 criteria for SUD	Regulation of Craving task with Drug and Food cues	Abstinent Condition vs. Non-abstinent Condition	fMRI	(1) Neurobiological Craving Signature (NCS): ventromedial prefrontal and cingulate cortices, ventral striatum, temporal/parietal association areas, mediodorsal thalamus and cerebellum (2) SUD showed a higher NCS response to drug cues than HC (p<0.001) (3) The NCS response to the drug versus food contrast discriminated between SUD and HC with an accuracy of 82% (p<0.001)	(1) Discrimination based on drug cues alone: NCS response classified SUD and HC with an accuracy of 75% based on drug cues alone (p=0.002311) (2) Cross-prediction using a food-cue-only pattern: Drug craving was also significantly predicted by a pattern that was trained only on food trials (p=0.004, AUC=0.74)
Van Timmeren et al. (2023)	AUD (n=46) HC (n=30) *AUD=Alcohol Using Disorder, HC=Healthy Controls *Diagnosed with DSM-5 criteria	Revised Monetary Incentive Delay Task (MIDT) with alcohol related cues	Between-group comparison	fMRI	(1) Reward Anticipation: AUD showed significantly decreased activity, relative to HC, while anticipating big versus small monetary rewards in three ROIs: the ventral striatum, putamen and caudate (all p<0.001) (2) Within AUD, significantly increased activation appeared in the bilateral precuneus extending to the middle cingulate, the bilateral frontal superior medial extending to the ACC, the left frontal inferior orbital cortex and several occipital regions (left angular and right lingual gyrus) (all p<0.05)	Cue Reactivity: No significant group difference in striatal activation was found between AUD and HC when comparing alcohol to neutral cues
Huang et al. (2024)	HUD (n=32) HC (n=21) *HUD=Heroin Use Disorder, HC=Healthy Controls *Diagnosed with DSM-5 criteria for opioid use disorder	Novel block-design fMRI cue reactivity task with drug, food and neutral cues	Between-group comparison	fMRI	Cue Reactivity - The HUD showed hyperactivation to drug cues(vs. neutral) compared to the HC in the left Nucleus Accumbens (NAcc) (extending to the subgenual ACC, vmPFC, the right Orbitofrontal Cortex (OFC) and the posterior Cingulate Cortex/Precuneus (p<0.001) - The HUD showed hyperactivation to drug cues(vs. food) compared to the HC in the right Inferior Frontal Gyrus and occipital regions compared to food cues (p<0.001)	(1) Within HUD, ventromedial prefrontal cortex (vmPFC) activity positively correlated with *drug craving (p<0.001) *Assessed by Heroin Craving Questionnaire scores (2) No significant group differences were found in the response to food cues compared to neutral cues (3) Reappraisal activity - Within the HUD, greater drug reappraisal activity in the IFG and dlPFC was associated with lower drug cue-induced craving (p<0.001)
Coquery et al. (2025)	Patients with obesity (n=40) No *FA (n=17) Low/Mild FA (n=10) High FA (n=13) *Diagnosed with Yale Food Addiction Scale (YFAS 2.0), Binge Eating Scale (BES), and Hospital Anxiety and Depression Scale (HADS)	Imaging task (1) Low Appetizing images (LA) (2) High Appetizing images (HA)	Between-group comparison	fMRI	(1) Higher FA was associated with lower brain response in the bilateral dorsal striatum (caudate and putamen) for HA but not for LA (p<0.001) (2) Higher FA correlated with lower brain responses in the pre-central gyrus for increasing hedonic level (LA to HA) (p<0.001) (3) FA severity is associated with impaired cortico-striatal functional connectivity (p<0.05): - i) the anterior cingulate cortex and two other ROIs: The left caudate and left accumbens; ii) the left accumbens and three other ROIs: anterior cingulate cortex, left insular cortex and left amygdala iii) the anterior cingulate cortex and the left thalamus (4) FA severity is associated with increased functional connectivity (p<0.05): - i) the left amygdala and the right hippocampus, ii) the left amygdala and the left hippocampus	For both LA and HA, similar patterns of activated clusters in brain regions were detected (p<0.001): In the gustatory and multisensory integration cortex (insula), caudate, putamen, middle cingulate cortex and thalamus
Wojciechowski et al. (2025)	*CSB (n=32) HC (n=31) *CSB=Compulsive Sexual Behaviors, HC=Healthy Controls *Diagnosed with Brief Pornography Screener (BPS), Sexual Addiction Screening Test (SAST-R), Criteria for Hypersexual Disorder (HD)	Active Appetitive Conditioning and Extinction	Between-group comparison	fMRI	Conditioning Phase (1) aOFC: Post-hoc testing showed increased activity in late conditioning in the CSB vs. HC (p<0.05) and an increase from the early to late conditioning phase in the CSB group (p<0.01) but not in the HC group, non-reward-specific (2) dACC: Post-hoc testing showed decreased activity in late conditioning in the CSB (p<0.05) but no change in the HC Extinction Phase (3) vStr: Post-hoc testing showed increased activity in late conditioning in the HC compared to CSB(p<0.05), non-reward-specific Extinction Phase (1) dACC: CSB showed generally higher activity compared to HC(p<0.05), non-reward-specific (2) vStr: In the late extinction phase, CSB showed a tendency for activity to decrease for sexual cues, but activity was maintained at a higher level for monetary cues than for sexual cues(t=3.41, p<0.01)	(1) CSB showed generally higher arousal and valence ratings for both sexual cues and monetary cues compared to the HC (2) No significant Group effect or interaction with Group effects was found in the Amygdala

Table 2

Authors (Year)	Samples (N)	Task	Contrast	Modality	Primary findings	Secondary findings
Meyer et al. (2004)	PG (n=14) HC (n=15) *PG=Problem Gamblers, HC=Healthy Controls *Diagnosis based on SOGS (South Oaks Gambling Screen) and KFG (Kurzfragebogen zum Glücksspielverhalten)	Blackjack game at the casino	Card game for accumulating points without financial interest	Endocrine measurements	(1) Elevated baseline epinephrine in PG compared with non-PG (p<0.001) (2) Higher Norepinephrine levels in PG compared with non-PG throughout the gambling session (p=0.007) (3) PG exhibited significantly higher Dopamine levels throughout the gambling session compared to non-PG (p=0.028)	(1) Actual gambling results positively correlated with Norepinephrine plasma concentrations (r=0.392, p<0.05) (2) Desire to gamble positively correlated with norepinephrine plasma concentrations (r=0.390, p<0.05)
Hollander et al. (2005)	PG (n=7) *PG=Pathological Gamblers *Diagnosis based on DSM-4 TR and SOGS (South Oaks Gambling Screen)	Online Blackjack game	Online Blackjack game without monetary reward	PET	(1) Significantly greater prefrontal activation during gambling for money (1.08 ± 0.19) compared with gambling for points alone (1.01 ± 0.23) (p=0.00062) with stronger effect in the right hemisphere compared to the left, and in anterior/dorsolateral areas compared to ventral areas (p=0.011) (2) Lower metabolic rate with money reward than no money in the Caudate, but the reverse in the Ventral putamen (p=0.03)	(1) Greater medial frontal cortex activation with money (1.10) compared with points alone (1.01) (p=0.046) (2) Greater activation in specific regions during the money reward condition positively correlated with higher *SOGS score - Visual cortex Brodmann area 17 (0.84), Brodman area 24 (left, 0.84), Brodmann area 11 (right, 0.92 left, 0.95) (p<0.05) *SOGS score = pathological gambling severity
De Ruiter et al. (2009)	PG (n=19) non-PG (n=19) Smokers (n=19) *PG=Problem Gamblers *Diagnosis based on DSM-4 TR	Probabilistic Reversal-Learning Task (PRLT)	Between-group comparison	fMRI	(1) Monetary gain: Significantly lower *VLPFC activation in PG relative to healthy controls (p<0.05) (2) Monetary loss: Significantly lower VLPFC activation in PG and Smokers relative to healthy controls (p<0.05) *VLPFC: Ventral Fronto-striatal Circuit	(1) Monetary gain: Significantly higher right insula activation in smoker relative to healthy controls (p<0.05)
Miedl et al. (2010)	PG (n=12) OG (n=12) *PG=Problem Gamblers, OG=Occasional Gamblers *Diagnosis based on DSM-4 and KFG (Kurzfragebogen zum Glücksspielverhalten)	Experimental blackjack task	Between-group comparison	fMRI	Win vs. Lose (reward processing) (1) Both groups showed common activation patterns in brain regions related to reward processing (Ncl.accumbens, bilateral frontal regions, left parietal regions (precuneus), left occipital regions, bilateral cerebellum, left thalamus, and right posterior cingulate gyrus) (p<0.05) (2) Significantly higher fronto-parietal activation (right superior frontal, left inferior parietal, left superior parietal) in PG relative to OG (p<0.001)	Low risk vs. High risk (risk assessment) (1) PG showed a greater activation increase in the high-risk compared to the low-risk in the right superior temporal gyrus, the right inferior frontal gyrus, and the right thalamus (p<0.001) (2) OG displayed the reverse activation pattern: activation increase in the low-risk compared to the high-risk (p<0.001)
van Holst et al. (2012)	PG (n=14) HC (n=15) *PG=Problem Gamblers, HC=Healthy Controls *Diagnosis based on DSM-4 TR and SOGS (South Oaks Gambling Screen)	Guessing task (1) Expectation (reward or loss) and anticipation phase (2) Outcome phase	Between-group comparison	fMRI	PG > HC (1) PG showed stronger activation in bilateral ventral striatum for larger reward amounts than for smaller reward amounts (p<0.05) (2) PG showed stronger activation related to *EV+ in bilateral dorsal striatum and left OFC (p<0.05) *EV+=Gain-Related Expected Value	Within the PG, *SOGS score showed a highly significant negative correlation (r=-0.76) with right amygdala activation related to *EV+ coding *SOGS score=the severity of gambling problems *EV+=Gain-Related Expected Value
Balodis et al. (2012)	PG (n=14) CC (n=14) *PG=Pathological Gambling, CC=Control Comparison *Diagnosis based on SOGS (South Oaks Gambling Screen)	Revised Monetary Incentive Delay Task (MID)	Between-group comparison	fMRI	The PG showed significantly reduced activity in the ventromedial prefrontal cortex (vmPFC), insula, and ventral striatum (VS) compared to the CC during several phases (p<0.05)	In the PG, negative correlation was observed between Ventral Striatum activity and measures of *impulsivity (p≤0.05) *Assessed by BIS-11 total score and subscale scores
Choi et al. (2012)	*PG (n=15) OCD (n=13) HC (n=15) *PG=Problem Gamblers, OCD=Obsessive-Compulsive Disorder, HC=Healthy Controls *Diagnosis based on DSM-4 (SCID) and SOGS (South Oaks Gambling Screen)	Monetary Incentive Delay task (MID)	Between-group comparison	fMRI	(1) The PG showed reduced neural activity in the ventromedial caudate nucleus during both gain anticipation and loss anticipation compared to OCD and HC (2) During gain anticipation, PG showed less activation in the ventromedial caudate nucleus, middle frontal gyrus, and occipital cortex compared to OCD (3) During loss anticipation, PG showed a markedly reduced ventromedial caudate nucleus activation compared to OCD and HC	(1) Within the PG, a significant positive correlation was found between anterior insula activity during loss anticipation and the *SOGS score (r=0.64, p=0.02) *SOGS=the severity of PG symptoms (2) The PG showed reduced thalamus activation during gain anticipation compared to OCD and HC (p<0.005)
Sescousse et al. (2013)	PG (n=20) HC (n=20) *PG=Problem Gamblers, HC=Healthy Controls *Diagnosis based on DSM-4 TR and SOGS (South Oaks Gambling Screen)	Incentive Delay Task with Monetary reward and Erotic reward (1) anticipation phase (2) outcome phase	Between-group comparison	fMRI	(1) Reward Anticipation Phase: The ventral striatum of PG showed a differential response to monetary cues versus erotic cues, which was primarily attributed to lower reactivity to erotic cues (p<0.001) (2) Reward Outcome Phase: - In HC, the posterior OFC primarily responded to erotic rewards - In PG, the posterior OFC region responded strongly to both monetary gains and erotic rewards (3) Hedonic Value Encoding in the ventral striatum No correlation with erotic reward ratings was observed in the PG	Differential response in ventral striatum was significantly positive correlated with the SOGS score (r=0.50) *SOGS score=the severity of gambling symptoms
Romanczuk et al. (2014)	*PG (n=18) AD (n=15) HC (n=17) *PG=Pathological Gambling, AD=Alcohol Dependent, HC=Healthy Controls *Diagnosis based on DSM-4 (SCID-I)	Monetary Incentive Delay (MID)	Between-group comparison	fMRI	(1) Loss Anticipation - The PG showed significantly stronger activation in the posterior region of the right ventral striatum compared to AD and HC (p<0.001) (2) Successful Loss Avoidance - The PG showed significantly reduced activation in the right ventral striatum and right medial prefrontal cortex compared to the HC (p<0.001) (3) Gain Anticipation and Outcome - PG did not show a significant difference in ventral striatum activity compared to the HC - AD showed significantly reduced activation in the left ventral striatum compared to the HC (p<0.001)	The recent *severity of gambling behavior showed a negative correlation with brain activation in the right medial prefrontal cortex (mPFC) during loss-aversion feedback *Assessed by PG-YBOCS
Bellmunt-Gil et al. (2025)	*GD (n=15) HC (n=17) *GD=Gambling Disorder, HC=Healthy Controls *Diagnosis based on DSM-4 and DSM-5	Visual task - visual stimuli: Gambling-related videos, Erotic videos and Neutral videos	Between-group comparison	fMRI PET	(1) GD showed significantly greater BOLD response in the dorsal striatum than HC when watching gambling-related versus neutral videos (p<0.05) (2) The BOLD response to gambling cues in the dorsal striatum and the ventral striatum showed a correlation in HC (r=0.7, p=0.003), but no correlation was found in GD (r=-0.1, p=0.75) (3) In GD, the BOLD response in the dorsal striatum to gambling cues showed a significant positive correlation with carfentanil BPND, which indicates mu-opioid receptor availability (r=0.81, p<0.001) but no correlation was found in HC (r=0.08, p=0.79)	No significant BOLD response difference when watching erotic and neutral videos

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